Together for Tomorrow!
Enabling People
Education for Future Generations

At-Risk Prediction: Forecasting Student Pass/Fail Outcomes

SIC – AI803

Agenda

- Team Members
- Project Definition
- Preprocessing
- EDA
- Model Selection
- Conclusion
- Future Enhancements & Recommendations

Team Members

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Problem Definition

Problem Definition

Identifying at-risk students before exam season

1 Context

Academic failure is often identified only after it has occurred. Our goal is to use information available before the exam season to identify students who may be at risk of failing and may require additional support.

2 Problem Statement

How can we use such information to identify students who are at risk of failing, and determine the main factors associated with it?

Before

Exam season the window for early intervention

Why

Support is targeted only after the factors behind failure are known

Who

We provide support to Students who are at risk of failing

Problem Objectives

What this project aims to deliver

1. **Early Risk Detection**

   Identify students who are at risk of failing before the exam season.

2. **Identify Main Contributing Factors**

   Determine which academic, behavioral, lifestyle, and other student factors are most correlated with failure.

3. **Actionable Support**

   Provide insights that help educational stakeholders take timely, targeted action to support at-risk students.

3
Core objectives

4
Factor categories studied

1
Unified goal: support at-risk students

Data Cleaning & Validation

Data Preprocessing

Student Exam Performance dataset - quality checks performed before modeling

✓ **No Duplicate Records**
0 duplicate rows and 0 duplicate student IDS across all 100,000.

0 **Clean target variable**
0 missing values in pass_status; Exactly two clases - pass and fail.

✓ **Plausible ranges**
Ages, scores, GPA, study hours, attendance and more all fall within realistic bounds.

⚠ **Fail = positive class**
Target set so identifying at-risk (failing) students is the primary objective.

100,000
Student records

44
Original features

22.6
Fail rate in data

Splitting, imputation & encoding

Preparing modern ready features while preventing data leakage

1 split
80/20 stratified split - done before imputation

→

2 Impute
Median (numeric) mode (categorical), from training data only

→

3 Encode
One hot including, drop_first=True

→

4 Scale
Standard scalar - scaled & unscaled versions kept

80,000 / 20,000
Train / test split

62
Model ready features

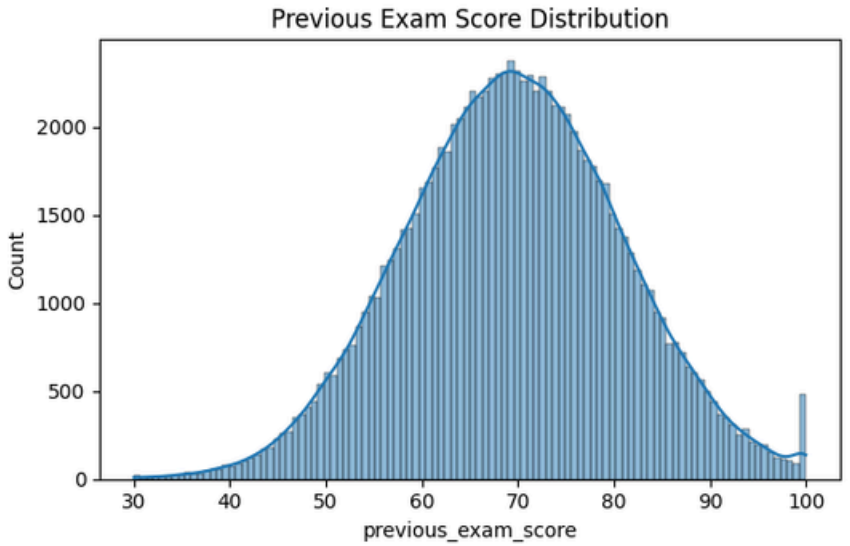
6
Leakage columns removed

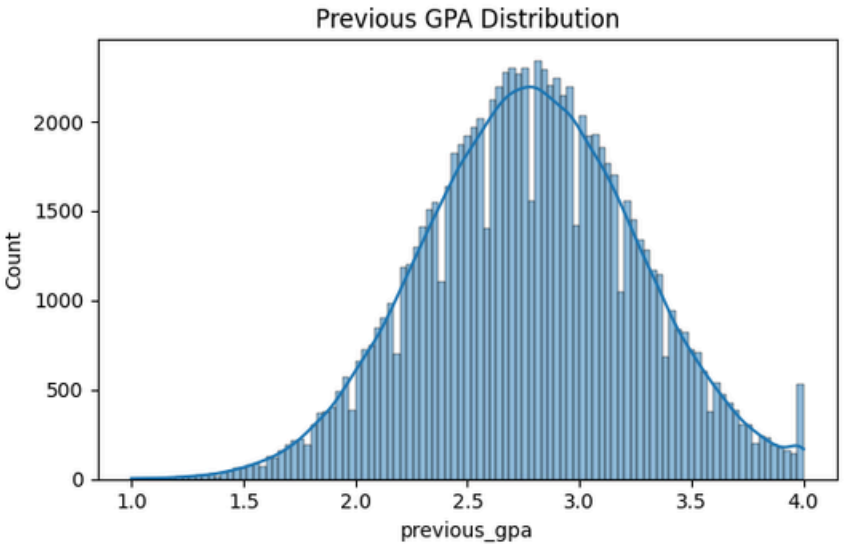
Removed to prevent data leakage: student_ID, questions_attempted, questions_correct, exam_score, performance_grade, performance_level.

EDA

Descriptive statistics

Central tendency & spread of key numerical variables





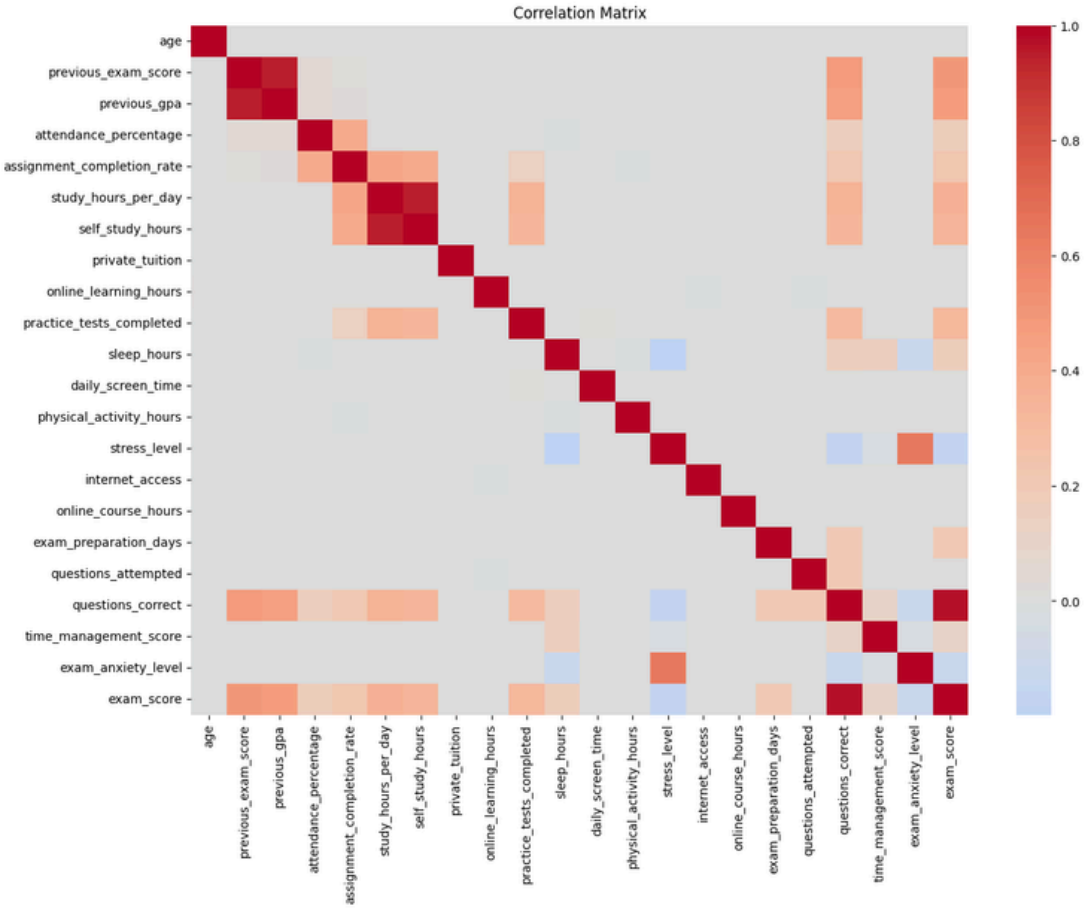
69.5 ± 11.4
Previous exam score

2.78 ± 0.48
Previous GPA

62.0± 15.9
Final exam score

Descriptive statistics

What variables move together, and what separates pass from fail



Strongest links

Previous_exam_score ↔ Previous_GPA, and questions_correct ↔ exam_score

Expected overlap

Study_hours_per_day ↔ self_study_hours (they measure similar things)

Mild negative

Stress_level ↔ sleep_hours

Key signal

Failing students cluster at lower GPA & lower previous exam scores

Selection of Model

Model Training & Evaluation

LR

Logistic Regression

Probability-based classifier

- Estimate Fail likelihood
- Support interpretable predictions
- Compare performance using consistent evaluation metrics

DT

Decision Tree

Rule-based nonlinear classifier

- Test nonlinear decision patterns
- Compare against the linear model
- Compare performance using consistent evaluation metrics

Model Training & Evaluation

RF

Random Forest

Ensemble tree-based classifier

- Combine multiple decision trees
- Reduce overfitting and improve robustness
- Compare performance using consistent evaluation metrics

GB

Gradient Boosting

Sequential boosting classifier

- Build trees sequentially to correct previous errors
- Improve predictive performance and accuracy
- Compare performance using consistent evaluation metrics

Model Training & Evaluation

EM

Ensemble Model

Combined Random Forest & XGBoost model

- Combine Random Forest and XGBoost predictions
- Improve stability and overall predictive performance
- Compare ensemble results against individual models

Feature Importance (Random Forest)

1

What it measures ?

How much each feature contributed to the model's predictions.

2

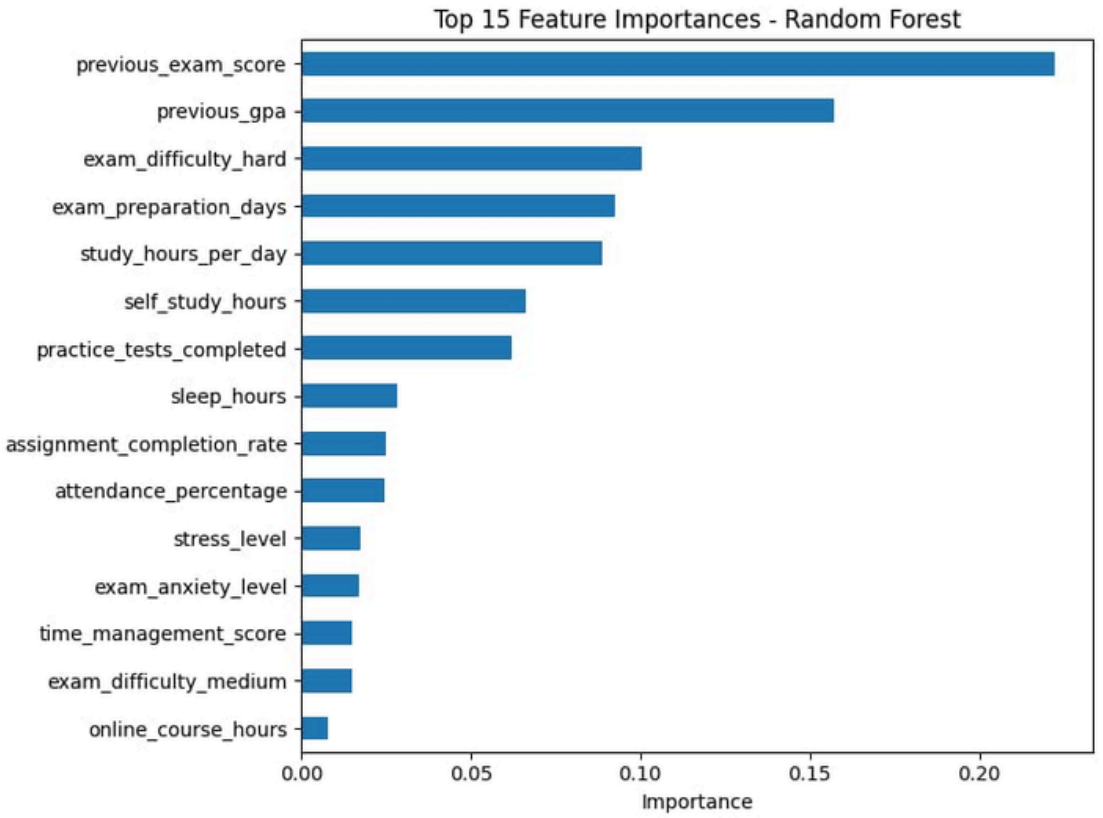
Top drivers

Previous_exam_score and previous_gpa are by far the strongest predictors (22% and 16%).

3

Why it matters

Confirms the model is learning real academic patterns, not noise - attendance and study habits also rank meaningfully.



Ensemble Model (Random Forest + XGBoost)

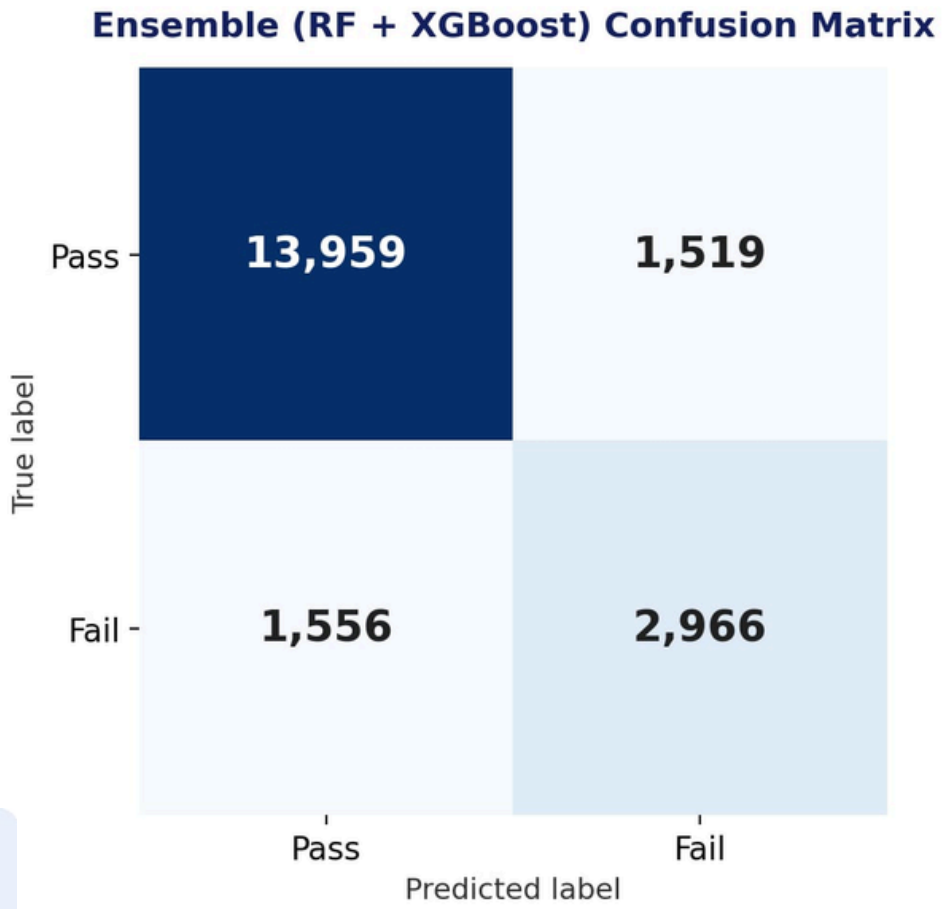
EM

Ensemble Model

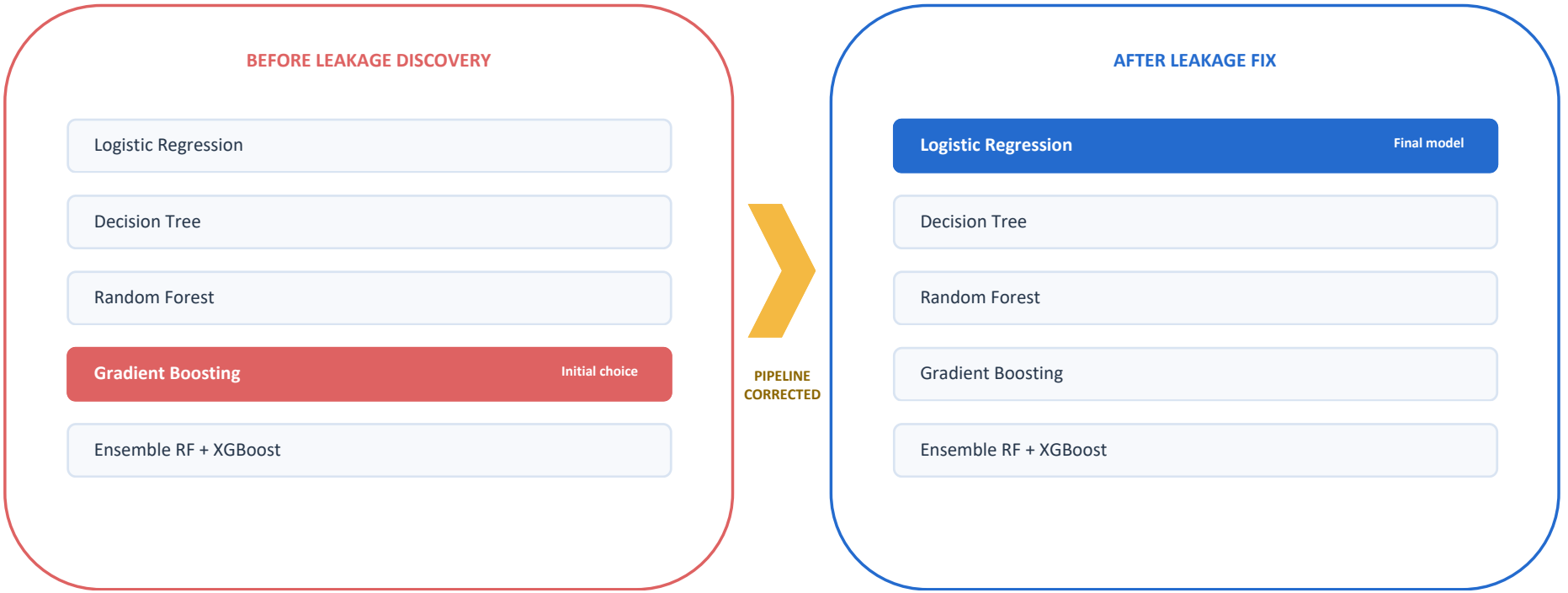
Combined Random Forest & XGBoost model

- Combines Random Forest and XGBoost predictions using soft voting
- Averages predicted probabilities from both models
- Reduces variance and improves stability vs. either model alone
- Accuracy improves to 84.6% (vs. 79.5% for Random Forest alone)

84.6%	66.1%	65.9%
Accuracy	Precision	F1-Score



Model Selection Before and After the Leakage Fix

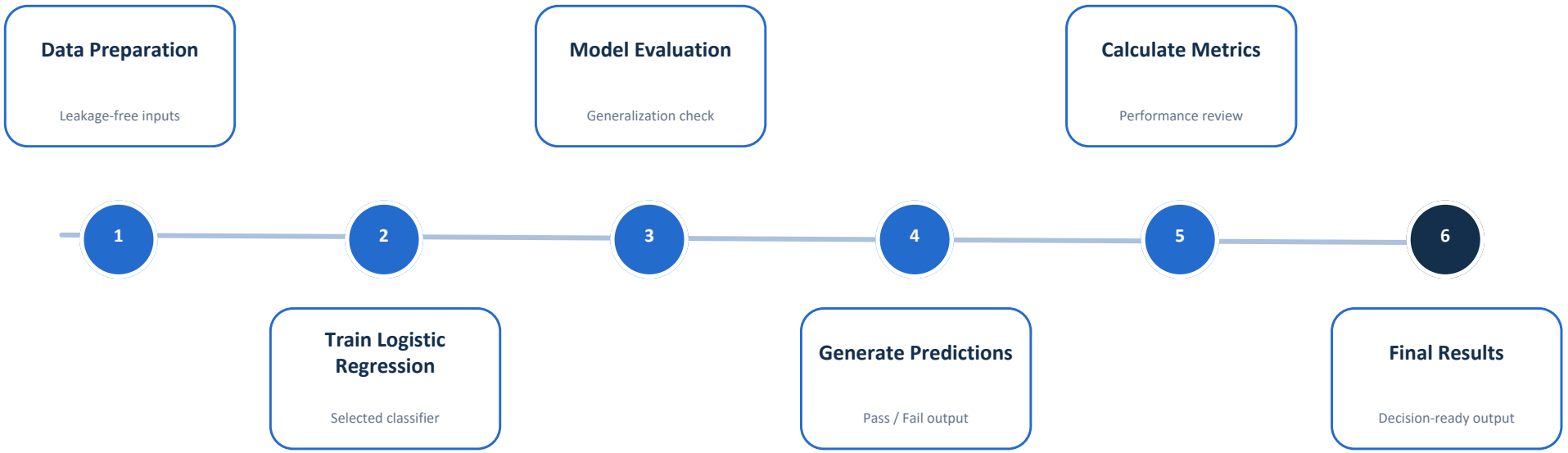


Final Model Comparison After the Leakage Fix

Model	Accuracy	Precision Fail	Recall Fail	F1 Fail
Logistic Regression	80%	53%	**82%**	65%
Decision Tree	77%	49%	79%	61%
Random Forest	79.50%	53.14%	78.88%	63.50%
Gradient Boosting	85.23%	71.83%	57.05%	63.59%
Ensemble RF + XGBoost	84.63%	66.13%	65.59%	65.90%

Logistic Regression was selected because it achieved the highest Fail recall (82%), detecting the largest share of at-risk students.

Work After Selecting Logistic Regression



Final Logistic Regression Results

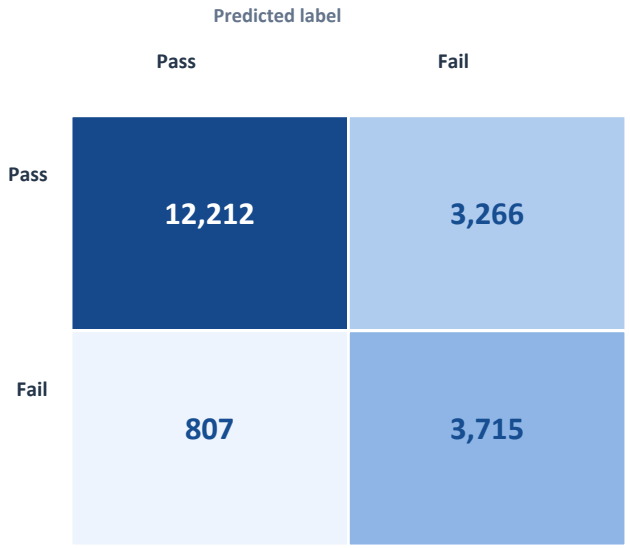
Accuracy	Precision - Fail	Recall - Fail
79.64%	53.22%	82.15%

F1 - Fail	ROC-AUC	PR-AUC
64.59%	0.89	0.7202

Classification Summary

Class	Precision	Recall	F1-score	Support
Pass	0.94	0.79	0.86	15,478
Fail	0.53	0.82	0.65	4,522

Confusion Matrix



Key result: 3,715 of 4,522 actual Fail cases were detected, producing 82.15% Fail recall.

Conclusion

Main Findings

5.1 What the analysis and modeling revealed

1

Previous Academic Performance

Previous exam score and previous GPA were the strongest predictors of future exam success — noticeably more important than the other variables.

2

Preparation & Study Behavior

Exam preparation days, study hours per day, self-study hours, and practice tests completed were also among the most important predictors — preparation patterns mattered.

Tuned LR

model selected due to high recall on the fail class

89.00%

ROC-AUC of the chosen model

Business Recommendations

5.2 Turning findings into action

1 Implement an Early Risk Identification System

Assign students a risk level based on their predicted probability of failing.

2 Focus on Weak Prior Academic Performance

Since previous exam score and GPA are the strongest predictors, flag students with consistently weak performance early for tutoring, revision sessions, or instructor guidance.

3 Encourage Structured Exam Preparation

Since prep days, study hours, and practice tests were important predictors, encourage earlier preparation with structured revision plans inspired by top preforming collegues.

Risk Tiers

Low Risk
Normal academic support

Medium Risk
Additional monitoring and academic guidance

High Risk
Immediate intervention and personalized support

Together for Tomorrow!

Enabling People

Education for Future Generations

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